

MULTI-COMPONENT SOUNDTRACK GENERATION WITH MULTIMODAL CONTROLS

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ABSTRACT

Recent multimodal-guided audio generation methods can synthesize audio well aligned with content and temporal dynamics. However, existing popular models still struggle to capture fine-grained content, with details such as speech content often insufficiently represented. In video-text-to-audio (VT2A) tasks, video provides general contextual information, while text contains explicit descriptions that are not always fully reflected. To this end, we propose MCST-Gen, a multi-component soundtrack generation model covering speech, sound effects (SFX), music and other components. We design a data processing pipeline that filters heterogeneous audio-visual data, identifies acoustic components, and organizes high-quality dataset with structured and component-aware annotations. We further incorporate large language model (LLM)-based text representation and curriculum learning strategy to enhance multimodal controllability. Experiments demonstrate that MCST-Gen produces coherent multi-component soundtrack while maintaining high-quality generation and precise alignment.

Index Terms— Multi-Component Soundtrack Generation, Multimodal Control, LLM-based Text Representation, Diffusion Transformer, Curriculum Learning

1. INTRODUCTION

Audio is an indispensable component of audiovisual content, complementing visual information through sound effects (SFX), speech, music and other acoustic events. This has driven the development of multimodal audio generation, with recent methods supporting diverse audio synthesis tasks from textual or visual inputs, including audio [1, 2, 3, 4, 5, 6], speech [7, 8, 9] and full-soundtrack generation [10, 11, 12, 13].

Despite these advances, existing video-text-to-audio (VT2A) models still face challenges in generating detailed audio content, particularly speech. In VT2A, video mainly provides general information, while text offers explicit content control [5], yet such details have not been fully exploited. MMAudio [4], AudioX [3], and HunyuanVideo-Foley [5] achieve strong semantic and temporal alignment under multimodal conditions, do not specifically target detailed textual control. UniFlow-Audio [10] supports diverse audio components but primarily models different generation tasks separately. Dasheng-AudioGen [11]

generates audio including speech and SFX but cannot perform video-conditioned audio generation. AudioGen-Omni [12] enables complex audio generation while using separate encoders for general and lyrics/transcription conditions, which may limit the consistency of semantic representations. Overall, effectively translating fine-grained textual information into concrete audio content while leveraging video context remains challenging.

To address this challenge, we propose MCST-Gen, a multi-component soundtrack generation framework covering speech, SFX and music controlled by multimodal conditions. We develop a data curation and restructuring pipeline for public text-audio and video-audio-text paired data, progressively filtering low-quality and weakly aligned samples, analyzing acoustic content, and organizing heterogeneous data into component-aware structured annotations. To better capture semantic and contextual information in the structured descriptions, we incorporate the representational capability of large language model (LLM) through Qwen3-8B-based text encoder [14], and introduce curriculum learning to be better controlled by multimodal guidance. Text and video conditions are fused with audio latents through multimodal Diffusion Transformers [15], followed by unimodal DiT refinement [16, 17], and optimized with flow matching [18] objective.

Experiments demonstrate that MCST-Gen effectively translates detailed multimodal conditions into coherent multi-component soundtracks with high quality, accurate semantic and temporal alignment. Our contributions are as follows: (1) we propose MCST-Gen, a multimodal controllable framework for multi-component soundtrack generation; (2) we design a data processing pipeline that filters, analyzes, and reorganizes multimodal datasets for component-aware structured annotations; (3) we integrate LLM-based representational ability through text encoder from Qwen3-8B; and (4) we leverage a curriculum learning strategy establish text-audio generation capabilities and progressively enhance multimodal alignment.

2. METHOD

2.1. Data Curation and Annotation

A key factor limiting the fine-grained controllability of existing audio generation models is that training data often contain low-quality segments and concise descriptions, providing insufficient supervision for individual components. To address these issues, we develop a multi-stage data curation and annotation pipeline that jointly controls signal quality and cross-modal rel-

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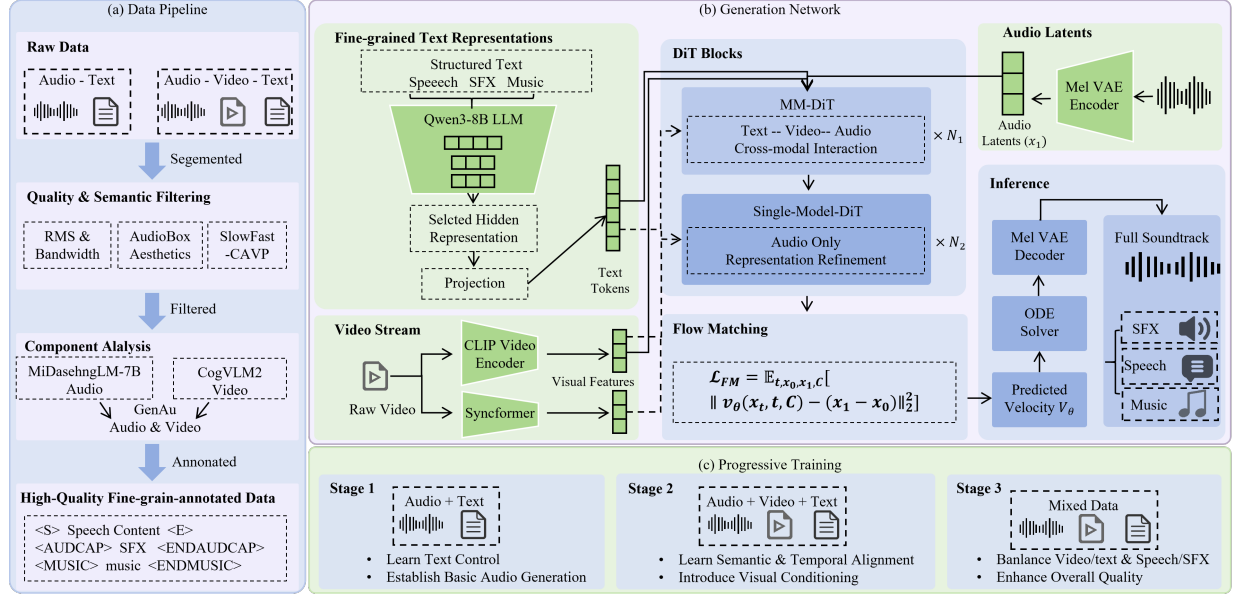


Fig. 1. Overview of MCST-Gen. (a) Data processing pipeline; (b) generation framework, where dashed lines denote auxiliary conditions and solid lines denote data or feature flows; (c) curriculum learning strategy.

evance, identifies acoustic components, and converts heterogeneous samples into a unified, component-aware training corpus, showed in in Fig. 1(a).

We first construct a hierarchical keyword set based on the AudioSet ontology [19] to collect publicly available audio-text and audiovisual data, which are then segmented into 10 s clips. We perform audio quality filtering using RMS-based silence detection ($RMS < 0.005$), effective bandwidth analysis, and AudioBox-Aesthetics [20], retaining clips with an AudioBox-Aesthetics score of at least 4. We further discard clips with weak audio-visual semantic correspondence according to SlowFast-CAVP [21], reducing cross-modal supervision noise. MiDashengLM-7B [22] is used to analyze and categorize the acoustic composition of the retained clips and route speech-containing samples to pretrained Mel-Band RoFormer denoising model [23] for speech-SFX separation, yielding cleaner component-level data for subsequent structured annotation.

Then we use MiDashengLM-7B [22] and CogVLM2 [24] for the retained clips to generate acoustic and visual descriptions. The two descriptions are fused into a unified cross-modal description using GenAU [25], and their semantic consistency is further checked to reduce cases where visually inferred events are absent from the actual audio. We apply CLAP-based [26] semantic ranking to identify potentially inconsistent annotations, followed by manual inspection and correction of low-consistency samples.

The resulting data are organized into a unified component-aware structured representation: $\langle S \rangle \dots \langle E \rangle$, $\langle \text{AUDCAP} \rangle \dots \langle \text{ENDAUDCAP} \rangle$, and $\langle \text{MUSIC} \rangle \dots \langle \text{ENDMUSIC} \rangle$, representing speech content, SFX, and music descriptions, respectively. Absent components are left empty. The content of each component is derived by consolidating relevant presence, semantic, and auxiliary attributes from the hierarchical annotations. All source datasets are mapped to this common representation, enabling consistent component-aware textual conditioning across heterogeneous data sources.

2.2. Model Architecture

The architecture of MCST-Gen is illustrated in Fig. 1(b). Following MMAudio [4], we build MCST-Gen upon a multimodal latent audio-generation architecture and adapt its conditioning interface for multi-component soundtrack generation. First, text, video, and audio are encoded by modality-specific encoders and projected into a common representation space. The resulting multimodal representations are jointly modeled by MM-DiT [15] blocks, followed by unimodal DiT [17] refinement with semantic and temporal conditioning. Finally, the refined audio latent is modeled by flow matching and decoded into the output waveform.

2.2.1. Multimodal Encoders

The component-level annotations obtained from data curation are serialized into a fixed structured prompt and encoded by an LLM-based text encoder through Qwen3-8B [14]. We extract token-level hidden states encoded by transformers of Qwen3-8B as the textual condition. These contextualized token representations preserve both local component descriptions and their relations within the structured prompt, enabling the model to preserve explicit semantics while retaining contextual relations among different components. Video frames are encoded by CLIP [27] to capture semantic visual information, while Synformer [28] provides higher-frame-rate visual features for temporal conditioning. The audio waveform is converted into a mel-spectrogram and encoded by a pretrained variational autoencoder (VAE) [29]. All modality features are converted into one-dimensional tokens and projected into a shared transformer dimension.

2.2.2. Diffusion Transformer

Given the encoded multimodal representations, the generation network comprises multimodal and audio-only DiT [17] blocks.

The MM-DiT jointly processes text, video, and audio tokens to capture cross-modal semantic dependencies, after which the audio-only DiT further refines the audio latent while focusing on its representation and temporal structure. In addition, semantic and high-frame-rate temporal features are injected through feature modulation, providing complementary guidance for audio content and audio-visual synchronization.

2.2.3. Flow Matching

We optimize the network with the conditional flow matching objective [18]. Given a target audio latent x_1 , a noise sample $x_0 \sim \mathcal{N}(0, I)$, and multimodal condition C , the model predicts the velocity of the linear interpolation $x_t = (1 - t)x_0 + tx_1$:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, x_0, x_1, C} \left[\|v_\theta(x_t, t, C) - (x_1 - x_0)\|_2^2 \right]. \quad (1)$$

At inference, we initialize the process with Gaussian noise and integrate the velocity field from $t = 0$ to $t = 1$ using Euler steps.

2.3. Training Strategy

We adopt a curriculum learning strategy shown in Fig. 1(c) to balance text controllability, audio quality, and audiovisual synchronization. Since video carries dense contextual and temporal information, jointly introducing text and video from the beginning may cause the model to tend to over-rely on visual cues and loses control from detailed textual conditions [6]. We therefore first train the model on audio-text pairs to establish a text-conditioned audio generation prior, and then introduce video-audio-text data to learn multimodal semantic and temporal correspondence. Finally, we perform mixed-data fine-tuning with a slightly adjusted sampling ratio to jointly optimize the complementary multimodal conditions and maintain a balanced learning of different generation capabilities. This staged training enables the model to progressively acquire and jointly optimize multimodal generation capabilities, reducing the risk of visual conditions dominating the detailed textual control.

3. EXPERIMENTS

3.1. Experimental Setup

Training.

The encoders are kept frozen, while our MCST-Gen is trained from scratch, enabling fixed-duration inference with text conditions and variable-duration inference with video conditions. We use the AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.95$, and a weight decay of 1×10^{-6} . The global batch size is 96, training on 8 GPUs. For each training stage, the learning rate is set on the order of 1×10^{-4} and decayed to 0.25 of its previous value at 50% and 80% of training. To support flexible combinations of input modalities, we randomly drop each model with a probability of 0.03 during training and use a CFG strength of 4.5 during inference.

Datasets. As described in Sec. 2.1, we reorganize approximately 1.3M samples, totaling around 3,000 hours of audio-text and video-audio-text data. The audio-text data include LibriTTS [30], LJSpeech [31], and AudioCaps [32], covering speech, general audio events, and sound effects. The video-audio-text data include ACAVCaps [33], VGGSound [34], OpenHumanVid [35], GRID [36], LRS2 [37], Hallo3, and

TED0521, providing visual context for speech, sound effects, and other acoustic components. All datasets are reorganized into a unified component-aware format and used throughout the progressive training procedure described above.

Evaluation. We evaluate the generated audio from several complementary aspects. Kullback–Leibler (KL) divergence computed from PaSST features measures the distributional similarity, while CLAP and ImageBind scores (IB) measure audio-text and audio-visual semantic alignment, respectively. AudioBox Aesthetics is used to assess production quality (PQ), content usefulness (CU), and content enjoyment (CE). We omit the production complexity (PC) score because it is difficult to interpret reliably for our multi-component soundtrack generation task. For speech, we report word error rate (WER) using Whisper-large-v3 [38], speaker similarity (Spk.Sim.) using Resemblyzer, and mel-cepstral distortion metrics including MCD and MCD-DS. LSE-D is used to measure audiovisual synchronization. Lower values are better for KL, WER, MCD, MCD-DS, and LSE-D, while higher values are better for the remaining metrics.

3.2. Main Results

We evaluate MCST-Gen on both audiovisual-text and audio-text paired data, covering single-component and full-soundtrack generation across video-to-audio (V2A), video-to-speech (V2S), video-to-soundtrack (V2ST), text-to-audio (TTA), text-to-speech (TTS) and text-to-music (TTM) tasks. For a fair comparison, all baselines use publicly released checkpoints with inputs converted to their required formats, while preserving the original conditioning information.

Video-conditioned generation. Table 1 summarizes video-conditioned generation across V2ST, V2A, and V2S settings. All baselines use publicly released checkpoints, with their inputs converted to the required formats while preserving the original conditioning information. We first evaluate soundtrack generation on OpenHuman and ACAVCaps, where the target soundtracks may contain speech, sound effects, and music. ACAVCaps uses its speech-containing subset for speech-related metrics, whereas OpenHuman consists entirely of speech-containing samples; therefore, MMAudio is not evaluated on OpenHuman. For both datasets, evaluation samples are randomly selected from the corresponding data distributions to preserve an in-distribution setting. On OpenHuman, MCST-Gen improves over Foley-Omni on semantic alignment, perceptual quality, speaker similarity, spectral distortion, and audio-visual synchronization, with only a modest increase in WER. On ACAVCaps, it achieves the best PQ, CU, speaker similarity, MCD, and MCD-DS, while other systems remain stronger on some distributional and semantic-alignment metrics. These results indicate that MCST-Gen can generate heterogeneous soundtracks while maintaining a favorable balance between overall audio quality and speech fidelity.

We further evaluate single-task generation on VGGSound and GRID, representing V2A and V2S, respectively. On VGGSound, MCST-Gen achieves the best PQ and CU among unified models and remains competitive with dedicated V2A systems, showing that unified multi-component modeling does not substantially compromise general audio generation. On GRID, it obtains the best KL and a WER close to that of the best dedicated V2S model, together with competitive IB, PQ, and CU scores. Dedicated V2S systems remain stronger on some speech-distortion and synchronization metrics, as expected from their task-specific design. WER should be inter-

Table 1. Visual-conditioned generation results.

Dataset	Type	Method	KL↓	IB↑	CLAP↑	PQ↑	CU↑	WER↓	Spk.Sim.↑	MCD↓	MCD-DS↓	LSE-D↓
VGGSound V2A	V2A	AudioX [3]	<u>1.770</u>	0.28	0.19	6.522	5.915	–	–	–	–	–
		HunyuanVideo-Foley [5]	<u>1.770</u>	<u>0.32</u>	<u>0.23</u>	<u>6.239</u>	<u>5.620</u>	–	–	–	–	–
		MMAudio [4]	1.410	0.34	0.35	6.058	5.506	–	–	–	–	–
	Unified	Uniflow-Audio [10]	1.976	0.30	0.23	<u>6.097</u>	5.510	–	–	–	–	–
		Foley-Omni [13]	1.632	<u>0.29</u>	0.29	6.087	<u>5.530</u>	–	–	–	–	–
		MCST-Gen(Ours)	<u>1.775</u>	0.27	<u>0.28</u>	6.272	5.741	–	–	–	–	–
GRID V2S GT WER:0.110	V2S	EmoDubber [8]	0.143	0.14	–	7.115	<u>6.635</u>	0.101	91.7	6.29	6.25	9.32
		ProDubber [9]	0.294	0.17	–	<u>6.752</u>	6.685	0.139	<u>89.7</u>	32.44	<u>6.45</u>	10.22
	Unified	Foleyomni [13]	<u>0.139</u>	0.15	–	5.302	4.923	0.145	83.8	<u>13.96</u>	8.14	<u>9.67</u>
		MCST-Gen(Ours)	0.093	<u>0.16</u>	–	6.556	6.220	<u>0.103</u>	85.2	15.12	8.05	9.77
	Unified	Foley-Omni [13]	<u>0.920</u>	<u>0.23</u>	<u>0.11</u>	<u>6.101</u>	<u>5.117</u>	0.220	<u>64.5</u>	16.99	<u>14.61</u>	<u>9.70</u>
		MCST-Gen(Ours)	0.805	0.36	0.18	6.620	5.606	<u>0.267</u>	65.1	15.16	12.73	9.68
OpenHumanvid V2ST	V2A	MMAudio [4]	1.404	0.28	0.22	<u>6.157</u>	5.239	1.139	<u>67.1</u>	18.64	16.34	8.77
		Foley-Omni [13]	1.635	<u>0.25</u>	<u>0.21</u>	6.040	<u>5.386</u>	0.266	67.0	<u>16.63</u>	<u>15.41</u>	9.54
	Unified	MCST-Gen(Ours)	<u>1.421</u>	0.24	0.19	6.202	5.453	<u>0.446</u>	68.6	15.71	14.46	<u>9.00</u>

Table 2. Textual-conditioned generation results.

Dataset	Type	Method	KL↓	CLAP↑	PQ↑	CE↑	CU↑	WER↓
AudioCaps TTA	TTA	DashengAudioGen	2.57	0.26	6.00	3.53	<u>5.42</u>	–
		MMAudio	3.00	0.31	5.90	3.30	5.29	–
		AudioX	1.47	<u>0.28</u>	5.85	3.57	5.10	–
	Unified	Foley-Omni	<u>2.43</u>	0.27	<u>5.99</u>	3.63	5.62	–
		MCST-Gen(Ours)	<u>2.43</u>	0.24	5.92	<u>3.59</u>	5.32	–
LibriTTS TTS GT WER:0.023	TTS	DashengAudioGen	0.75	–	6.32	<u>5.10</u>	<u>5.77</u>	0.129
		Foley-Omni	0.57	–	5.80	5.03	5.46	0.118
	Unified	MCST-Gen(Ours)	0.25	–	<u>6.05</u>	5.15	5.81	<u>0.124</u>
	TTM	MMAudio	<u>1.61</u>	0.14	6.48	5.45	6.21	–
		DashengAudioGen	2.16	0.15	6.39	4.84	5.66	–
ACAVCaps Music subset TTM	Unified	Foley-Omni	1.71	<u>0.15</u>	6.21	<u>5.57</u>	<u>6.27</u>	–
		MCST-Gen(Ours)	1.21	0.23	6.97	5.65	6.74	–

Table 3. Ablation of text representation and training strategy.

Setting	Variant	KL↓	IB↑	CLAP↑	WER↓	MCD↓	LSE-D↓
Text Encoder	T5-XXL	<u>0.82</u>	<u>0.32</u>	0.34	1.17	–	–
	Qwen3-Embedding	0.80	0.33	0.34	<u>1.01</u>	–	–
	Qwen3-8B	<u>0.82</u>	0.30	0.31	0.83	–	–
Training Strategy	Full Model	0.83	0.36	0.18	0.27	15.16	9.68
	w/o Fine-tuning	0.88	<u>0.35</u>	<u>0.17</u>	<u>0.27</u>	16.39	9.76
	w/o Curriculum Learning	0.89	0.33	0.16	0.43	16.85	9.68

interpreted with caution, since ASR systems may favor particular characteristics of generated speech; thus, a WER lower than that of the ground-truth reference does not necessarily indicate more accurate speech content.

Text-conditioned generation. We further evaluate text-conditioned generation on AudioCaps, LibriTTS, and the music-containing subset of ACAVCaps, as shown in Table 2. On AudioCaps, MCST-Gen remains competitive with Foley-Omni in KL and several quality-related metrics, although its CLAP score is lower than those of several baselines. On LibriTTS, it achieves the best KL, CE, and CU while maintaining competitive PQ and WER, indicating that unified modeling preserves strong text-to-speech capability. On the ACAVCaps music subset, MCST-Gen achieves the best KL, CLAP, PQ, CE,

and CU, indicating promising text-to-music performance and semantic alignment. Overall, MCST-Gen maintains competitive performance across individual tasks while supporting unified generation of different acoustic components.

3.3. Ablation Studies

We conduct ablation experiments to investigate the effects of text representation and training strategy. We compare three text encoders, Qwen3-Embedding [39], T5-XXL [40], and Qwen3-8B [14], under a lightweight training setting, and further evaluate MCST-Gen without the mixed-data fine-tuning stage and without curriculum learning.

As shown in Table 3, Qwen3-8B achieves the best WER among the three encoders, indicating that its LLM-based representation is more effective for preserving fine-grained speech content. Qwen3-Embedding and T5-XXL obtain better results on some general audio and semantic metrics, suggesting that the richer Qwen3-8B representation may require more training to fully benefit these metrics. For the training strategy, removing either the mixed-data fine-tuning stage or curriculum learning degrades most metrics, while the full model achieves the best overall performance. This demonstrates the importance of both curriculum learning and multimodal adaptation for fine-grained soundtrack generation.

4. CONCLUSION

In this work, we present MCST-Gen, a unified framework for multi-component soundtrack generation under multimodal controls. We develop a data processing pipeline to filter, analyze and reorganize open-source data, providing richer supervision for the joint generation of speech, SFX and music with structured annotations. We further incorporate the textual representation capability of LLM and adopt curriculum learning strategy to strengthen the control of fine-grained multimodal information during generation. Given video and structured textual conditions, MCST-Gen can generate coherent full-soundtrack while maintaining high audio quality with accurate semantic and temporal alignment.

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